15 · Case 10: stroke, and the map of the brain

The words of this track, as it says them. Bold brown is the examiner.

Exam case 10. Stroke and the brain.

You're working in a stroke rehabilitation unit, often with people recovering from a stroke, or cerebrovascular accident, where parts of the brain were damaged during periods of ischaemia. Different areas of the brain control different tasks, so the functions a person has lost reflect the part of the brain that was damaged. You're working with people with a range of losses, and thinking about which parts of the brain must be involved.

Two big questions, seven marks each, on her cerebrum diagram. Her sessions say the exam uses the same letters. Then the brain stem questions from her formative test.

Seven marks. Using the image, name the functional area of the cerebrum at each letter, A to G.

The hook, from A to G: plan, speak, move, feel, hear, understand, see.

Letter A, the prefrontal cortex. Plan.

B, Broca's area. Speak.

C, the motor area, on the precentral gyrus. Move.

D, the somatosensory area, on the postcentral gyrus. Feel.

E, the auditory area. Hear.

F, Wernicke's area. Understand.

G, the occipital area, the visual cortex. See.

Seven marks. For each region, what changes to function or behaviour could follow damage to it?

Lose the area, and you lose its job.

Area A, the prefrontal cortex, does problem-solving, judgement, holding back inappropriate behaviour, and keeping the mind on goals. Damage it, and you get poor judgement, inappropriate behaviour, and a loss of focus.

Area B, Broca's area, produces speech and writing that others can understand. Damage gives difficulty producing speech and writing.

Area C, the motor area, instructs the muscle movements, with every muscle group mapped. Damage gives a loss of voluntary movement in the mapped parts.

Area D, the somatosensory area, receives touch and the other general senses, from every part of the body. Damage gives a loss of touch and general sensation.

Area E, the auditory area, is for hearing, with regions tuned to different frequencies. Damage affects hearing.

Area F, Wernicke's area, understands spoken and written language. Damage means the person can't understand language.

Area G, the occipital area, is the visual processing system, the primary visual cortex. Damage affects vision.

One honest note. Her deck gives what each area does. The "damage leads to" half is the podcast's reading of each function. It's the obvious reading, but it isn't her slide.

Her formative test asks four of these as drop-downs. Judgement, reasoning and impulse control: area A, the prefrontal cortex. Producing speech others can understand: area B, Broca's area. Making sense of sounds: area E, the primary auditory cortex. Controlling voluntary movement: area C, the primary motor cortex.

And the brain stem, from her formative test. Four people, four injuries.

A person who took no alcohol or drugs behaves like a drunk person after a car accident, and the blood alcohol test comes back negative. The damaged part: the cerebellum.

After a head injury, someone breathes with long, gasping in-breaths, a pause at full inspiration, and a short out-breath. That's the pons, because the apneustic centre there controls the length of inspiration.

After falling down the stairs, someone breathes very slowly and shallowly, and responds poorly to high carbon dioxide or low oxygen. That's the medulla oblongata, because it contains the pacemaker of the respiratory system.

After a motorbike accident, someone's left and right hands act independently. She can't coordinate tasks that need both hands, like tying shoelaces, and one hand acts in conflict with the other: the alien hand. That's the corpus callosum.

Then she asks where each one sits on her diagram, as fill-in-the-blanks. Here are her keys. The cerebellum is directly posterior to the fourth ventricle, directly inferior to the inferior colliculus, and directly superior to the reticular formation. The pons is on the anterior side of the brain stem, directly inferior to the midbrain, where the inferior colliculus is labelled, directly superior to the medulla oblongata, and directly anterior to the fourth ventricle. The medulla oblongata is directly anterior to the cerebellum, and directly inferior to the pons. And the corpus callosum is directly superior to the lateral ventricle at the front, and directly superior to the fornix at the back. It's inferior to the cingulate gyrus of the cerebrum.

That's thirteen blanks on her test. Positions are hard to learn by ear alone, so look at her diagram once with these words playing.

Once more. Case 10.

Area A, the prefrontal cortex: judgement, so damage brings poor judgement and inappropriate behaviour. B, Broca's area: producing speech. C, the motor area, on the

precentral gyrus: voluntary movement. D, the somatosensory area, on the postcentral gyrus: touch. E, the auditory area: hearing. F, Wernicke's area: understanding language. G, the occipital area: vision. Lose the area, lose its job. Drunk with no alcohol: the cerebellum. Long, gasping in-breaths: the pons. Very slow, shallow breathing: the medulla oblongata. Hands acting independently: the corpus callosum.